# Telecom Customer Churn Prediction

End-to-End Data Analytics Project — Final Report

Python (pandas) | SQL | Power BI

## 1. Introduction

Customer churn — when a customer stops using a company's services — is one of the most critical challenges faced by telecom companies. Acquiring a new customer typically costs significantly more than retaining an existing one, which makes early identification of at-risk customers a high-value business problem. This project analyzes historical customer data from a telecom company (7,043 customers) to understand churn behavior and uncover the key factors that drive customers to leave.

## 2. Objectives

- Clean and prepare the raw customer dataset using Python (pandas).
- Perform in-depth business analysis using SQL to answer key churn-related questions.
- Design an interactive Power BI dashboard to visualize churn patterns for business stakeholders.
- Translate the findings into clear, actionable retention recommendations.

## 3. Dataset

The project uses the IBM Telco Customer Churn dataset — 7,043 customers and 21 attributes, covering demographics (gender, senior citizen, partner, dependents), account details (tenure, contract type, payment method, billing), subscribed services (phone, internet, online security, tech support, streaming, etc.), and the churn label.

## 4. Tools & Technology

**Python (pandas)** was used for data cleaning and feature engineering. **SQL** was used for structured business analysis and answering key churn questions. **Power BI** was used to build an interactive churn-insights dashboard.

## 5. Data Cleaning (Python / pandas)

The raw dataset was cleaned and prepared in Python before analysis. The steps followed were:

- **Column names standardized** — all column names converted to lowercase, readable snake_case (e.g. MonthlyCharges → monthly_charges).
- **Missing values fixed** — the total_charges column had blank values for new customers (tenure = 0); these were converted to numeric and filled with 0.
- **Data types corrected** — total_charges converted from text to numeric; categorical columns (gender, contract, churn, etc.) converted to category type.
- **Unnecessary column removed** — customer_id was dropped before analysis since it holds no predictive/analytical value.
- **New features engineered** — tenure_group (tenure bucketed into 0-12, 13-24, 25-48, 49-60, 60+ months), avg_monthly_spend (total_charges / tenure), num_services (count of subscribed add-on services), and is_new_customer (flag for tenure < 6 months) were added to support deeper analysis.
- **Duplicate check** — dataset was checked for duplicate rows/customer IDs; none were found.

## 6. SQL Analysis — Business Questions & Findings

The cleaned dataset was queried using SQL to answer the following business questions. Key results are summarized below.

### Q1: Total Number of Customers

| Total Customers |
|---|
| 7,043 |

### Q2: Overall Churn Rate

Out of 7,043 total customers, 1,869 have churned and 5,174 are retained.

| Churn | Customer Count | Percentage |
|---|---|---|
| Yes | 1,869 | 26.54% |
| No | 5,174 | 73.46% |

### Q3: Churn Rate by Contract Type

| Contract | Total Customers | Churned | Churn Rate |
|---|---|---|---|
| Month-to-month | 3,875 | 1,655 | 42% |
| One year | 1,473 | 166 | 11% |
| Two year | 1,695 | 48 | 2% |

Month-to-month contract customers churn far more than annual/biennial contract customers.

### Q4: Average Tenure — Churned vs Retained

| Churn | Avg. Tenure (months) | Customer Count |
|---|---|---|
| Yes | 17 | 1,869 |
| No | 37 | 5,174 |

### Q5: Churn Rate by Payment Method

| Payment Method | Total Customers | Churned | Churn Rate |
|---|---|---|---|
| Electronic check | 2,365 | 1,071 | 45% |
| Mailed check | 1,612 | 308 | 19% |
| Bank transfer (automatic) | 1,544 | 258 | 16% |
| Credit card (automatic) | 1,522 | 232 | 15% |

Electronic check users churn at nearly 3x the rate of automatic payment users.

### Q6: Churn Rate — Senior Citizens

| Senior Citizen | Total Customers | Churned | Churn Rate |
|---|---|---|---|
| Yes | 1,142 | 476 | 41% |
| No | 5,901 | 1,393 | 23% |

### Q7: Top 10 Customers by Monthly Charges

| Customer ID | Monthly Charges | Tenure | Contract | Churn |
|---|---|---|---|---|
| 8199-ZLLSA | $118.35 | 67 | One year | Yes |

| Customer ID | Monthly Charges | Tenure | Contract | Churn |
|---|---|---|---|---|
| 2889-FPWRM | $117.80 | 72 | One year | Yes |
| 2302-ANTDP | $117.45 | 48 | Month-to-month | Yes |
| 9053-JZFKV | $116.20 | 67 | Two year | Yes |
| 1444-VVSGW | $115.65 | 70 | One year | Yes |
| 0201-OAMXR | $115.55 | 70 | One year | Yes |
| 4361-BKAXE | $114.50 | 41 | Month-to-month | Yes |
| 1555-DJEQW | $114.20 | 70 | Two year | Yes |
| 9158-VCTQB | $113.60 | 41 | Month-to-month | Yes |
| 7279-BUYWN | $113.20 | 41 | Month-to-month | Yes |

Notably, all of the top 10 highest-paying customers in the dataset have churned — reinforcing that high-value customers are a critical retention priority.

### Q8: Churn Rate by Internet Service

| Internet Service | Total Customers | Churned | Churn Rate | Avg. Monthly Charges |
|---|---|---|---|---|
| Fiber optic | 3,096 | 1,297 | 41.89% | $91.5 |
| DSL | 2,421 | 459 | 18.96% | $58.1 |
| No internet service | 1,526 | 113 | 7.4% | $21.08 |

Fiber optic customers churn the most, despite paying the highest average monthly charges — a potential pricing or service-quality issue.

### Q9: Churn Rate by Number of Services

| Service Group | Total Customers | Churned | Churn Rate |
|---|---|---|---|
| Zero services | 80 | 35 | 43.75% |
| One or more services | 6,963 | 1,834 | 26.34% |

### Q10: Churn Rate by Tenure Group

| Tenure Group (months) | Total Customers | Churned | Churn Rate |
|---|---|---|---|
| 0–12 | 2,186 | 1,037 | 47% |
| 13–24 | 1,024 | 294 | 28% |
| 25–48 | 1,594 | 325 | 20% |
| 49–60 | 832 | 120 | 14% |
| 60+ | 1,407 | 93 | 6% |

Churn risk drops sharply as tenure increases — the first 12 months are the most critical retention window.

## Q11: Churn Rate by Paperless Billing

| Paperless Billing | Total Customers | Churned | Churn Rate |
|---|---|---|---|
| Yes | 4,171 | 1,400 | 33% |
| No | 2,872 | 469 | 16% |

## Q12: Average Monthly Charges — Churned vs Retained

| Churn | Avg. Monthly Charges | Customer Count |
|---|---|---|
| Yes | $74.44 | 1,869 |
| No | $61.27 | 5,174 |

## Q13: Contract Type + Internet Service (Combined)

| Contract | Internet Service | Total | Churned | Churn Rate |
|---|---|---|---|---|
| Month-to-month | Fiber optic | 2,128 | 1,162 | 54% |
| Month-to-month | DSL | 1,223 | 394 | 32% |
| One year | Fiber optic | 539 | 104 | 19% |
| Month-to-month | No internet | 524 | 99 | 18% |
| One year | DSL | 570 | 53 | 9% |
| Two year | Fiber optic | 429 | 31 | 7% |
| One year | No internet | 364 | 9 | 2% |
| Two year | DSL | 628 | 12 | 1% |
| Two year | No internet | 638 | 5 | 0% |

Month-to-month + Fiber optic is the single highest-risk customer segment, with a 54% churn rate.

## Q14: Churn Rate by Multiple Lines

| Multiple Lines | Total Customers | % of Customers | Churned | Churn Rate |
|---|---|---|---|---|
| Yes | 2,971 | 42% | 850 | 28% |
| No | 3,390 | 48% | 849 | 25% |
| No phone service | 682 | 9% | 170 | 24% |

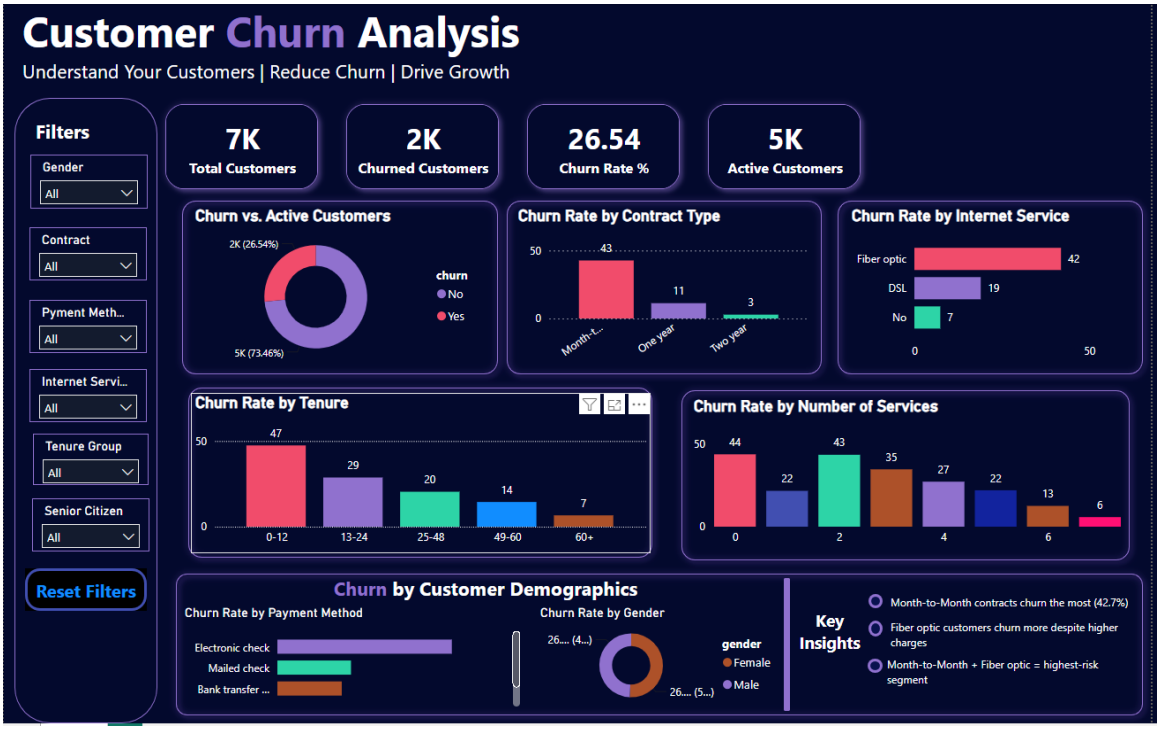
## Q15: Churn Rate by Partner / Dependents

| Partner | Dependents | Total Customers | Churned | Churn Rate |
|---|---|---|---|---|
| No | No | 3,280 | 1,123 | 34% |
| Yes | No | 1,653 | 420 | 25% |
| No | Yes | 361 | 77 | 21% |
| Yes | Yes | 1,749 | 249 | 14% |

Customers with neither a partner nor dependents churn the most — customers with both churn the least.

## 7. Power BI Dashboard

An interactive Power BI dashboard was built on top of the cleaned dataset and the SQL analysis above, allowing stakeholders to explore churn patterns by contract type, internet service, tenure, number of services, payment method, and demographics — with slicers for gender, contract, payment method, internet service, tenure group, and senior citizen status.




Customer Churn Analysis Dashboard — Power BI

## 8. Key Insights

- Month-to-Month contracts churn the most (42.7%).
- Fiber optic customers churn more despite paying higher charges.
- The first 12 months of tenure carry the highest churn risk (47%).
- Electronic Check users churn nearly 3x more than automatic-payment users.
- Senior citizens churn more than non-seniors (41% vs 23%).
- Customers with zero add-on services churn more than those with multiple services.
- Month-to-Month + Fiber optic is the single highest-risk segment (54% churn rate).

## 9. Recommendations

- Incentivize month-to-month customers to switch to annual/two-year contracts through targeted discounts.
- Investigate service quality and pricing for Fiber optic plans, given their disproportionately high churn.
- Focus retention efforts on customers in their first 12 months — the highest-risk window.
- Encourage a shift away from Electronic check payments toward automatic payment methods.
- Promote add-on services (security, backup, tech support) to increase customer engagement and stickiness.

## 10. Conclusion

This project analyzed 7,043 telecom customer records using Python for data cleaning, SQL for structured business analysis, and Power BI for interactive visualization. The analysis identified contract type, tenure,

internet service type, and payment method as the strongest drivers of churn, with the Month-to-Month + Fiber optic segment being the single highest-risk group. These insights provide a clear, data-driven foundation for the business to design targeted customer-retention strategies.